

Inequalities

- **Definition 1**

Let $a, b \in \mathbb{R}$ with $a < b$. We define the sets:

$$(a, b) = \{x \in \mathbb{R} : a < x < b\}$$

is called the **open interval** between a and b .

$$[a, b] = \{x \in \mathbb{R} : a \leq x \leq b\}$$

is called the **closed interval** between a and b .

$$(a, b] = \{x \in \mathbb{R} : a < x \leq b\}$$

is called the **half-open interval** between a and b .

$$[a, b) = \{x \in \mathbb{R} : a \leq x < b\}$$

is called the **half-open interval** between a and b .

$$(a, \infty) = \{x \in \mathbb{R} : x > a\}$$

is the open interval of all numbers greater than a .

$$(-\infty, b) = \{x \in \mathbb{R} : x < b\}$$

is the open interval of all numbers smaller than b .

$$[a, \infty) = \{x \in \mathbb{R} : x \geq a\}$$

is the half-open interval of all numbers greater than or equal to a .

$$(-\infty, b] = \{x \in \mathbb{R} : x \leq b\}$$

is the half-open interval of all numbers smaller than or equal to b .

We also have

$$(-\infty, \infty) = \mathbb{R}.$$

- **Remark 1.1**

The symbols $-\infty$ and ∞ are not real numbers; they indicate that the corresponding interval has no endpoint.

If $a = b$, then we define $[a, b] = \{a\}$ and $(a, b) = \emptyset$.

- **Example 1.2**

The conditions

$$-2 < x \leq 3, \quad x < 1 \vee x \geq 4, \quad 2 \leq x \leq 5$$

describe the sets

$$(-2, 3], \quad (-\infty, 1) \cup [4, \infty), \quad [2, 5].$$

- **Example 1.3**

Prove that for every $x \in \mathbb{R}$, if $x^3 + x > 0$ then $x > 0$.

- **Proof 1.3.1**

If $x^3 + x > 0$, then $x(x^2 + 1) > 0$. Since $x^2 + 1 > 0$, we have $(x^2 + 1)^{-1} > 0$. Hence

$$x(x^2 + 1) \cdot (x^2 + 1)^{-1} > 0 \cdot (x^2 + 1)^{-1}$$

$$x > 0.$$

◦ **Example 1.4**

Solve the inequality

$$\frac{x+2}{x+4} > \frac{x-3}{2x-1}.$$

■ **Solution 1.4.1**

$$\begin{aligned} \frac{x+2}{x+4} > \frac{x-3}{2x-1} &\Leftrightarrow \frac{x+2}{x+4} - \frac{x-3}{2x-1} > 0 \\ &\Leftrightarrow \frac{(x+2)(2x-1) - (x-3)(x+4)}{(x+4)(2x-1)} > 0 \\ &\Leftrightarrow \frac{2x^2 + 3x - 2 - x^2 - x + 12}{(x+4)(2x-1)} > 0 \\ &\Leftrightarrow \frac{x^2 + 2x + 10}{(x+4)(2x-1)} > 0 \\ &\Leftrightarrow \frac{(x+1)^2 + 9}{(x+4)(2x-1)} > 0. \end{aligned}$$

Since $(x+1)^2 + 9 > 0$, the sign of $\frac{(x+1)^2 + 9}{(x+4)(2x-1)}$ depends on the sign of $(x+4)(2x-1)$. Consider the points where the factors are zero, -4 and $\frac{1}{2}$, which divide the set of real numbers into three intervals, that is, $(-\infty, -4)$, $(-4, \frac{1}{2})$, $(\frac{1}{2}, \infty)$. Then we have:

	$x+4$	$2x-1$	$(x+4)(2x-1)$
$x < -4$	−	−	+
$-4 < x < \frac{1}{2}$	+	−	−
$x > \frac{1}{2}$	+	+	+

Therefore, the solution is $(-\infty, -4) \cup (\frac{1}{2}, \infty)$.

◦ **Theorem 1.5**

If $a \in \mathbb{R}$ such that $0 \leq a < \varepsilon$ for every $\varepsilon > 0$, then $a = 0$.

■ **Proof 1.5.1**

If $a > 0$, then let $\varepsilon = \frac{a}{2} > 0$. By the hypothesis, we have that

$$0 \leq a < \frac{a}{2}.$$

But $a > \frac{a}{2}$ since $a > 0$, which is a contradiction. Therefore, $a = 0$.

◦ **Notation 1.6**

If $a \in \mathbb{R}$ with $a \neq 0$, $a^{-1} = \frac{1}{a}$.

◦ **Theorem 1.7**

Let $a, b \in \mathbb{R}$.

1. If $a > 0$, then $a^{-1} > 0$.
2. If $0 < a < b$, then $a^{-1} > b^{-1}$.

■ **Proof 1.7.1**

1. Suppose that $a > 0$ and $a^{-1} < 0$, then

$$1 = aa^{-1} < 0,$$

which is a contradiction. Therefore, $a^{-1} > 0$.

2. Suppose that $0 < a < b$. Then by Theorem 1.7(1), we have that $a^{-1} > 0$ and $b^{-1} > 0$. Hence

$$\begin{aligned} a &< b \\ a \cdot a^{-1} &< b \cdot a^{-1} \\ 1 &< ba^{-1} \\ b^{-1} \cdot 1 &< b^{-1} \cdot ba^{-1} \\ b^{-1} &< a^{-1}. \end{aligned}$$

• **Definition 2**

The **absolute value** of $a \in \mathbb{R}$ is defined by

$$|a| = \begin{cases} a & (a \geq 0) \\ -a & (a < 0) \end{cases}.$$

◦ **Proposition 2.1**

Let $a \in \mathbb{R}$, then:

1. $|a| \geq 0$.
2. $-|a| \leq a \leq |a|$.
3. $|a| = \sqrt{a^2}$.
4. $|ab| = |a||b|$.
5. $|-a| = |a|$.

■ **Proof 2.1.1**

1. If $a \geq 0$, then $|a| = a \geq 0$. If $a < 0$, then $|a| = -a > 0$.
2. If $a \geq 0$, then $|a| = a$ and thus

$$-|a| = -a \leq 0 \leq a = |a|.$$

If $a < 0$, then $|a| = -a$ and thus

$$-|a| = a \leq 0 \leq -a = |a|,$$

since $-a > 0 > a$. Therefore, $-|a| \leq a \leq |a|$ holds for all cases.

3. Since $\sqrt{a^2}$ is the unique nonnegative real number such that

$$\left(\sqrt{a^2}\right)^2 = a^2,$$

we have that $\sqrt{a^2} = |a|$ or $\sqrt{a^2} = -|a|$. But by definition, $\sqrt{a^2}$ is a nonnegative real number, thus $\sqrt{a^2} = |a|$.

4. By the properties of exponents and the square root of positive real numbers, we have:

$$\begin{aligned} |ab| &= \sqrt{(ab)^2} \\ &= \sqrt{a^2b^2} \\ &= \sqrt{a^2}\sqrt{b^2} \\ &= |a||b|. \end{aligned}$$

5. Since $(-a)^2 = a^2$, by Property 3 we have:

$$|-a| = \sqrt{(-a)^2} = \sqrt{a^2} = |a|.$$

◦ **Remark 2.2**

$$|a|^2 = |a^2| = a^2 \text{ since } a^2 \geq 0.$$

◦ **Proposition 2.3**

Let $a, b \in \mathbb{R}$. Prove that:

1. If $c \geq 0$, then $|a| \leq c \iff -c \leq a \leq c$.
2. If $c \geq 0$, then $|a| \geq c$ if and only if $a \leq -c \vee a \geq c$.

■ **Proof 2.3.1**

1. ($\implies$) Suppose that $|a| \leq c$, then $-c \leq -|a|$. Together with Property 2, we have that

$$-c \leq -|a| \leq a \leq |a| \leq c,$$

that is, $-c \leq a \leq c$.

($\impliedby$) Suppose that $-c \leq a \leq c$. If $a \geq 0$, then $|a| = a$ and thus $|a| = a \leq c$. If $a < 0$, then $|a| = -a$. Since $-c \leq a \implies -a \leq c$, we have $|a| = -a \leq c$. Thus, in both cases, $|a| \leq c$.

2. ($\implies$) Suppose that $|a| \geq c$. If $a \geq 0$, then $a = |a| \geq c$. If $a < 0$, then $-a = |a| \geq c$, that is, $a \leq -c$.

($\impliedby$) Suppose that $a \leq -c \vee a \geq c$. If $a \leq -c$, then $-a \geq c \geq 0$ and thus $|a| = -a \geq c$. If $a \geq c \geq 0$, then $|a| = a \geq c$.

◦ **Example 2.4**

Solve $|5x + 2| \geq 4$.

■ **Solution 2.4.1**

$$\begin{aligned} |5x + 2| \geq 4 &\iff 5x + 2 \leq -4 \vee 5x + 2 \geq 4 \\ &\iff 5x \leq -6 \vee 5x \geq 2 \\ &\iff x \leq -\frac{6}{5} \vee x \geq \frac{2}{5}. \end{aligned}$$

Therefore, the solution is the set $(-\infty, -\frac{6}{5}] \cup [\frac{2}{5}, \infty)$.

◦ **Theorem 2.5 (Triangle inequality)**

For all $a, b \in \mathbb{R}$

$$|a + b| \leq |a| + |b|.$$

■ **Proof 2.5.1**

Since $ab \leq |ab| = |a||b|$ then

$$\begin{aligned} |a + b|^2 &= |(a + b)^2| \\ &= (a + b)^2 \\ &= a^2 + 2ab + b^2 \\ &\leq |a|^2 + 2|a||b| + |b|^2 \\ &= (|a| + |b|)^2 \end{aligned}$$

Hence $|a + b| \leq |a| + |b|$.